

Ali Nikkhah

Open to Relocation & Remote | alinkkh9@gmail.com | +98 991 296 3951 | alinikkhah2001.github.io |
linkedin.com/in/alinikkhah2001

SUMMARY

AI Engineer with 2+ years of experience designing and deploying production agentic systems, LLM-powered backends, and intelligent retrieval infrastructure. At Digikala, led the enterprise AI intelligence layer serving 5M+ daily queries, transitioning from static semantic matching to stateful LangGraph multi-agent pipelines with hybrid RAG (FAISS + Elasticsearch via RRF), HyDE query expansion, and async FastAPI serving backed by vLLM and Triton. At Turquoise Digital, engineered a financial agentic chatbot with real-time voice transcription (Whisper, wav2vec2.0) and dynamic function-calling for secure live portfolio queries. Strong foundation across the full stack: MLOps, distributed LLM inference, and high-concurrency backend systems.

TECHNICAL SKILLS

Languages & Runtimes: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, R, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, TensorFlow, JAX, Hugging Face Transformers, Time-Series Transformers, ONNX
Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, KAG, Graph RAG, FAISS, ChromaDB, Pinecone
Vision & Audio: OpenCV, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, ESPnet, TensorRT, TensorFlow Lite
Data Engineering: Apache Spark (PySpark), Trino, ClickHouse, Airflow, Kafka, PeerDB, Elasticsearch, Parquet, Delta Lake
MLOps & Serving: Ray, Triton Inference Server, vLLM (PagedAttention), TensorFlow Serving, TensorRT-LLM, MLflow, W&B, DVC
DevOps & Infrastructure: Docker, Kubernetes, Terraform, ArgoCD, GitHub Actions, Jenkins, GitLab CI
Databases & Monitoring: PostgreSQL, Redis, Prometheus, Grafana, ELK Stack, Metabase, Great Expectations

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

AI Engineering R&D Lead

Jan 2025 – Aug 2025

- Led the full architectural transition of the enterprise search and knowledge layer from static semantic matching to stateful multi-agent pipelines powered by **LangGraph**; the system now handles over **5 million daily queries** across product discovery, customer support, and internal knowledge retrieval.
- Designed a Modular Control Pipeline (MCP) that cleanly decouples text ingestion, chunking, retrieval, and response synthesis phases, enabling live LLM hot-swapping and zero-downtime vector index migrations without service interruption.
- Built a hybrid retrieval engine combining dense **FAISS** vector search with sparse **Elasticsearch** keyword lookup, fused via Reciprocal Rank Fusion (RRF); implemented **HyDE** query expansion and hierarchical chunking strategies to significantly close the accuracy gap on long-tail and ambiguous product queries.
- Deployed **Knowledge-Augmented Generation (KAG)** alongside standard RAG to enforce symbolic reasoning constraints on structured product catalog data, substantially reducing factual hallucinations in generated customer-facing answers.
- Engineered async **FastAPI** serving layers with multi-tier **Redis** caching and asynchronous request batching; deployed LLMs via **vLLM** with PagedAttention and **Triton Inference Server** to maximize KV-cache utilization and GPU throughput under burst traffic conditions.

Digikala

Tehran, Iran

AI Engineer

Nov 2024 – Jan 2025

- Prototyped and benchmarked RAG architectures against real production query logs; integrated LangGraph cyclic agent graphs for multi-turn session management, dynamic context re-evaluation, and automated query rewriting loops.
- Evaluated retrieval precision and end-to-end latency across candidate retrieval systems; findings directly shaped the R&D roadmap adopted in the subsequent lead role.

Turquoise Digital (Firouzeh Digital)

Tehran, Iran

Senior Data Engineer & Data Scientist

Sep 2025 – Mar 2026

- Engineered an enterprise **LangGraph** agentic chatbot handling customer support ticketing, financial literacy instruction, and live portfolio inquiries; integrated real-time **Whisper** and **wav2vec2.0** voice transcription so call-center audio streams feed the agent directly, eliminating manual ticket creation.
- Implemented strict data-access validation layers allowing the agent to safely execute compound-profit calculations via dynamic function-calling and retrieve live user equity balances, without ever exposing raw database credentials to the LLM context window.

- Built the AI portfolio allocation system: ingested user risk-tolerance questionnaires, fused historical time-series pricing data with streaming news-sentiment embeddings, and trained time-series transformer models to produce portfolio value forecasts with explicit statistical confidence intervals feeding automated trading allocations.

HomaCloud

Tehran, Iran

MLOps Engineer

Aug 2021 – Feb 2022

- Orchestrated distributed model training pipelines on Kubernetes using Ray for workload scheduling and MLflow for experiment tracking and artifact lineage.
- Built model inference APIs in FastAPI backed by TensorFlow Serving; automated full deployment lifecycle with Docker, ArgoCD, and GitHub Actions, cutting release cycles to under 10 minutes.

SELECTED PROJECTS & RESEARCH

AI Portfolio Allocation & Risk Engine

Turquoise Digital

Core Architect

2025 – 2026

- Designed an end-to-end asset recommendation system that processes user risk-tolerance questionnaires, fuses historical price time-series with real-time market-sentiment news embeddings, and applies time-series transformer architectures to project future portfolio valuations with statistical confidence intervals.
- Connected model outputs to automated batch trading execution, with explicit guardrails validating order size against computed risk limits before any submission.

Forex Heuristic Market Prediction Engine

Personal Research

Quantitative Developer

2024

- Formalized professional trader chart-reading heuristics — dynamic trendlines, support/resistance zones, and market microstructure signals — into structured feature matrices consumable by deep sequence models.
- Trained sequence-to-sequence networks on these derived representations to detect trend reversals and rank trade execution signals, treating expert visual pattern recognition as a learnable inductive bias.

RESEARCH HIGHLIGHTS

University of British Columbia

Remote Research Collaboration

Research Assistant – Vision-Language Medical Imaging

Feb 2024 – Nov 2025

- Developed vision-language architectures combining **ViT**, **BLIP**, and **T5** under self-supervised contrastive learning to automate generation of DICOM-compliant radiology reports from raw ultrasound images.
- Achieved state-of-the-art **BLEU** and **ROUGE-L** scores on the evaluation benchmark; preprint in preparation.

L3S Research Center (Leibniz University Hannover)

Remote Research Collaboration

Research Assistant – Video Motion Classification

Apr 2023 – Feb 2024

- Designed texture-free motion classification pipelines using 3D CNNs and transformer encoders over dense optical flow fields, deliberately suppressing appearance cues so models learn generalizable human-dynamic representations.
- Achieved state-of-the-art accuracy on **UCF-101** and **HMDB-51** benchmarks.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering – Digital Systems GPA: 3.65 overall, 3.91 major

Sept 2020 – Jul 2024